

This R code sample comes from a semester-long statistics course group project where we modeled global tree deforestation using environmental, urbanization, and consumption-related predictors from a credible open-source dataset. I led the modeling process and R programming, including building candidate linear regression models, applying backward selection, and comparing performance using 10-fold cross-validation with MAE and RMSE metrics. I focused on identifying a final model that was both statistically robust and easy to interpret, and helped create clear, user-friendly visualizations to communicate the key findings in the final presentation to an audience of 80+ people.

```
```{r, include = FALSE}
#full model
lm.func = function(data_clean){
 mod=lm(TreeLoss~ .,data=data_clean)
 return (mod)
}

crossed=na.omit(data_clean) %>%
 crossv_kfold(10)

crossed_1 = crossed %>%
 mutate(tr.model=map(train,lm.func))

crossed_2 = crossed_1 %>%
 mutate(predict=map2(test,tr.model,~augment(.y,newdata=.x))) %>%
 select(predict) %>%
 unnest()

MAE.func=function(residual){
 mae=mean(abs(residual),na.rm=T)
 return(mae)
}

RMSE.func=function(residual){
 rmse=sqrt(mean((residual)^2,na.rm=T))
 return(rmse)
}

full.MAE = MAE.func(crossed_2$.resid)
full.RMSE = RMSE.func(crossed_2$.resid)

...

```{r, include = FALSE}
`Cleaned P-value Model - 6
Variables`=lm(TreeLoss~Co2emissions+Meatsum+Forest_Tonsum+Forest_Volumesum+UrbanPop+`Urban%`,data = data_clean)
```

```

`CO2 Model - 1`=lm(Tree loss~ Co2emissions,data=data_clean)

`Meatsum - 1`=lm(Tree loss~ Meatsum,data=data_clean)

`Forest_ton - 1`=lm(Tree loss~ Forest_Tonsum,data=data_clean)

`Forest_vol - 1`=lm(Tree loss~ Forest_Volumesum,data=data_clean)

`Urban_pop - 1`=lm(Tree loss~ UrbanPop,data=data_clean)

`Urban_% - 1`=lm(Tree loss~ `Urban%`,data=data_clean)

#`Three Predictor Model` = lm(Forest_Volumesum+UrbanPop+`Urban%`,data = data_clean)
...

```{r, include = FALSE}
#mod3
lm.func = function(data_clean){
 mod=lm(Tree loss~
Co2emissions+Meatsum+Forest_Tonsum+Forest_Volumesum+UrbanPop+`Urban%`,data =
data_clean)
 return (mod)
}

crossed=na.omit(data_clean) %>% crossv_kfold(10)

crossed_1 = crossed %>%
 mutate(tr.model=map(train,lm.func))

crossed_2 = crossed_1 %>%
 mutate(predict=map2(test,tr.model,~augment(.y,newdata=.x))) %>%
 select(predict) %>%
 unnest()

MAE.func=function(residual){
 mae=mean(abs(residual),na.rm=T)
 return(mae)
}

RMSE.func=function(residual){
 rmse=sqrt(mean((residual)^2,na.rm=T))
 return(rmse)
}

mod3.MAE = MAE.func(crossed_2$.resid)
mod3.RMSE = RMSE.func(crossed_2$.resid)

...

```

```

```{r, include = FALSE}
#mod4
lm.func = function(data_clean){
  mod=lm(Tree loss~ Co2emissions,data=data_clean)
  return (mod)
}

crossed=na.omit(data_clean) %>% crossv_kfold(10)

crossed_1 = crossed %>%
  mutate(tr.model=map(train,lm.func))

crossed_2 = crossed_1 %>%
  mutate(predict=map2(test,tr.model,~augment(.y,newdata=.x))) %>%
  select(predict) %>%
  unnest()

MAE.func=function(residual){
  mae=mean(abs(residual),na.rm=T)
  return(mae)
}

RMSE.func=function(residual){
  rmse=sqrt(mean((residual)^2,na.rm=T))
  return(rmse)
}

mod4.MAE = MAE.func(crossed_2$.resid)
mod4.RMSE = RMSE.func(crossed_2$.resid)

...

```{r, include = FALSE}
#mod5
lm.func = function(data_clean){
 mod=lm(Tree loss~ Meatsum,data=data_clean)
 return (mod)
}

crossed=na.omit(data_clean) %>% crossv_kfold(10)

crossed_1 = crossed %>%
 mutate(tr.model=map(train,lm.func))

crossed_2 = crossed_1 %>%
 mutate(predict=map2(test,tr.model,~augment(.y,newdata=.x))) %>%

```

```

 select(predict) %>%
 unnest()

MAE.func=function(residual){
 mae=mean(abs(residual),na.rm=T)
 return(mae)
}

RMSE.func=function(residual){
 rmse=sqrt(mean((residual)^2,na.rm=T))
 return(rmse)
}

mod5.MAE = MAE.func(crossed_2$.resid)
mod5.RMSE = RMSE.func(crossed_2$.resid)
...

```{r, include = FALSE}
#mod6
lm.func = function(data_clean){
  mod=lm(TreeLoss~ Forest_Tonsum,data=data_clean)
  return (mod)
}

crossed=na.omit(data_clean) %>% crossv_kfold(10)

crossed_1 = crossed %>%
  mutate(tr.model=map(train,lm.func))

crossed_2 = crossed_1 %>%
  mutate(predict=map2(test,tr.model,~augment(.y,newdata=.x))) %>%
  select(predict) %>%
  unnest()

MAE.func=function(residual){
  mae=mean(abs(residual),na.rm=T)
  return(mae)
}

RMSE.func=function(residual){
  rmse=sqrt(mean((residual)^2,na.rm=T))
  return(rmse)
}

mod6.MAE = MAE.func(crossed_2$.resid)
mod6.RMSE = RMSE.func(crossed_2$.resid)
...

```

```

```{r, include = FALSE}
#mod7
lm.func = function(data_clean){
 mod=lm(Tree loss~ Forest_Volumesum,data=data_clean)
 return (mod)
}

crossed=na.omit(data_clean) %>% crossv_kfold(10)

crossed_1 = crossed %>%
 mutate(tr.model=map(train,lm.func))

crossed_2 = crossed_1 %>%
 mutate(predict=map2(test,tr.model,~augment(.y,newdata=.x))) %>%
 select(predict) %>%
 unnest()

MAE.func=function(residual){
 mae=mean(abs(residual),na.rm=T)
 return(mae)
}

RMSE.func=function(residual){
 rmse=sqrt(mean((residual)^2,na.rm=T))
 return(rmse)
}

mod7.MAE = MAE.func(crossed_2$.resid)
mod7.RMSE = RMSE.func(crossed_2$.resid)
...

```{r, include = FALSE}
#mod8
lm.func = function(data_clean){
  mod=lm(Tree loss~ UrbanPop,data=data_clean)
  return (mod)
}

crossed=na.omit(data_clean) %>% crossv_kfold(10)

crossed_1 = crossed %>%
  mutate(tr.model=map(train,lm.func))

crossed_2 = crossed_1 %>%
  mutate(predict=map2(test,tr.model,~augment(.y,newdata=.x))) %>%
  select(predict) %>%
  unnest()

```

```

MAE.func=function(residual){
  mae=mean(abs(residual),na.rm=T)
  return(mae)
}

RMSE.func=function(residual){
  rmse=sqrt(mean((residual)^2,na.rm=T))
  return(rmse)
}

mod8.MAE = MAE.func(crossed_2$resid)
mod8.RMSE = RMSE.func(crossed_2$resid)
...

```{r, include = FALSE}
#mod9
lm.func = function(data_clean){
 mod=lm(TreeLoss~ `Urban%`,data=data_clean)
 return (mod)
}

crossed=na.omit(data_clean) %>% crossv_kfold(10)

crossed_1 = crossed %>%
 mutate(tr.model=map(train,lm.func))

crossed_2 = crossed_1 %>%
 mutate(predict=map2(test,tr.model,~augment(.y,newdata=.x))) %>%
 select(predict) %>%
 unnest()

MAE.func=function(residual){
 mae=mean(abs(residual),na.rm=T)
 return(mae)
}

RMSE.func=function(residual){
 rmse=sqrt(mean((residual)^2,na.rm=T))
 return(rmse)
}

mod9.MAE = MAE.func(crossed_2$resid)
mod9.RMSE = RMSE.func(crossed_2$resid)
...

```{r, include = FALSE}
# Construct a table with RMSE & MAE values

```

```

vector1 <- c("Full Model - 8 Variables", "Cleaned P-value Model - 6 Variables", "CO2 Model",
"Meatsum Model", "Forest_ton Model", "Forest_vol Model", "Urban_pop Model", "Urban_%
Model")
vector2 <- c(full.MAE, mod3.MAE, mod4.MAE, mod5.MAE, mod6.MAE, mod7.MAE,
mod8.MAE, mod9.MAE)
vector3 <- c(full.RMSE, mod3.RMSE, mod4.RMSE, mod5.RMSE, mod6.RMSE,
mod7.RMSE, mod8.RMSE, mod9.RMSE)

combined_table <- tibble(
  Model = vector1,
  MAE = vector2,
  RMSE = vector3
)

...

```{r, include = FALSE}
#Backward elimination
data_clean <- na.omit(model_data)
Fit the full model
lm.model=lm(Tree loss~ .,data=data_clean)
Find the RMSE for the full model
RMSE = (summary(lm.model)$sigma)^2
Backward: use the step() function starting with the full model
step(lm.model,scale = RMSE)
...

```{r, include = FALSE}
mod_new2=lm(Tree loss~
Co2emissions+Meatsum+Forest_Tonsum+Forest_Volumesum+UrbanPop+`Urban%`,data =
data_clean)

mod_int=lm(Tree loss~
Co2emissions+Meatsum+Forest_Tonsum+Forest_Volumesum+UrbanPop+`Urban%` +
I(Co2emissions*Meatsum)+I(Co2emissions*UrbanPop)+I(Co2emissions*Forest_Tonsum),da
ta = data_clean)

...

```{r, include = FALSE}
lm.func = function(data_clean){
 mod=lm(Tree loss~
Co2emissions+Meatsum+Forest_Tonsum+Forest_Volumesum+UrbanPop+`Urban%` +
I(Co2emissions*Meatsum)+ I(Co2emissions*UrbanPop)
+I(Co2emissions*Forest_Tonsum),data = data_clean)
 return (mod)
}

```

```

crossed=na.omit(data_clean) %>% crossv_kfold(10)

crossed_1 = crossed %>%
 mutate(tr.model=map(train,lm.func))

crossed_2 = crossed_1 %>%
 mutate(predict=map2(test,tr.model,~augment(.y,newdata=.x))) %>%
 select(predict) %>%
 unnest()

MAE.func=function(residual){
 mae=mean(abs(residual),na.rm=T)
 return(mae)
}

RMSE.func=function(residual){
 rmse=sqrt(mean((residual)^2,na.rm=T))
 return(rmse)
}

mod_int.MAE = MAE.func(crossed_2$resid)
mod_int.RMSE = RMSE.func(crossed_2$resid)
mod_int.MAE
mod_int.RMSE

...

```{r, include = FALSE}
# int 1 mse & mae
`Interaction Model 1` = lm(TreeLoss~
Co2emissions+Meatsum+Forest_Tonsum+Forest_Volumesum+UrbanPop+`Urban%` +
I(Co2emissions*Meatsum),data=data_clean)

lm.func = function(data_clean){
  mod=lm(TreeLoss~
Co2emissions+Meatsum+Forest_Tonsum+Forest_Volumesum+UrbanPop+`Urban%` +
I(Co2emissions*Meatsum),data = data_clean)
  return (mod)
}

crossed=na.omit(data_clean) %>% crossv_kfold(10)

crossed_1 = crossed %>%
  mutate(tr.model=map(train,lm.func))

crossed_2 = crossed_1 %>%
  mutate(predict=map2(test,tr.model,~augment(.y,newdata=.x))) %>%
  select(predict) %>%

```



```

    unnest()

MAE.func=function(residual){
  mae=mean(abs(residual),na.rm=T)
  return(mae)
}

RMSE.func=function(residual){
  rmse=sqrt(mean((residual)^2,na.rm=T))
  return(rmse)
}

mod_int1.MAE = MAE.func(crossed_2$.resid)
mod_int1.RMSE = RMSE.func(crossed_2$.resid)

...

```{r, include = FALSE}
int 2 mse & mae
`Interaction Model 2` = lm(TreeLoss~
Co2emissions+Meatsum+Forest_Tonsum+Forest_Volumesum+UrbanPop+`Urban%`+
I(Co2emissions*Meatsum)+ I(Co2emissions*UrbanPop),data=data_clean)

lm.func = function(data_clean){
 mod=lm(TreeLoss~
Co2emissions+Meatsum+Forest_Tonsum+Forest_Volumesum+UrbanPop+`Urban%`+
I(Co2emissions*Meatsum)+ I(Co2emissions*UrbanPop),data = data_clean)
 return (mod)
}

crossed=na.omit(data_clean) %>% crossv_kfold(10)

crossed_1 = crossed %>%
 mutate(tr.model=map(train,lm.func))

crossed_2 = crossed_1 %>%
 mutate(predict=map2(test,tr.model,~augment(.y,newdata=.x))) %>%
 select(predict) %>%
 unnest()

MAE.func=function(residual){
 mae=mean(abs(residual),na.rm=T)
 return(mae)
}

RMSE.func=function(residual){
 rmse=sqrt(mean((residual)^2,na.rm=T))
 return(rmse)
}

```

```

}

mod_int2.MAE = MAE.func(crossed_2$.resid)
mod_int2.RMSE = RMSE.func(crossed_2$.resid)
...

```{r, include = FALSE}
# int 3 rmse & mae
`Interaction Model 3` = lm(TreeLoss~
Co2emissions+Meatsum+Forest_Tonsum+Forest_Volumesum+UrbanPop+`Urban%`+
I(Co2emissions*Meatsum)+ I(Co2emissions*UrbanPop)
+I(Co2emissions*Forest_Tonsum),data=data_clean)

lm.func = function(data_clean){
  mod=lm(TreeLoss~
Co2emissions+Meatsum+Forest_Tonsum+Forest_Volumesum+UrbanPop+`Urban%`+
I(Co2emissions*Meatsum)+ I(Co2emissions*UrbanPop)
+I(Co2emissions*Forest_Tonsum),data = data_clean)
  return (mod)
}

crossed=na.omit(data_clean) %>% crossv_kfold(10)

crossed_1 = crossed %>%
  mutate(tr.model=map(train,lm.func))

crossed_2 = crossed_1 %>%
  mutate(predict=map2(test,tr.model,~augment(.y,newdata=.x))) %>%
  select(predict) %>%
  unnest()

MAE.func=function(residual){
  mae=mean(abs(residual),na.rm=T)
  return(mae)
}

RMSE.func=function(residual){
  rmse=sqrt(mean((residual)^2,na.rm=T))
  return(rmse)
}

mod_int3.MAE = MAE.func(crossed_2$.resid)
mod_int3.RMSE = RMSE.func(crossed_2$.resid)
...

```